

Unit II :: Work and Energy

 **PHY01MX01R04: Mechanics and Properties of Matter** 

Dr. Parag Bhattacharya
Department of Physics
Rangapara College (Autonomous)

Syllabus

Unit II: Work and Energy

Concepts of work, power and energy; conservative and non-conservative forces; the work-energy theorem; conservative force as the negative gradient of potential energy; the law of conservation of mechanical energy; work done by non-conservative forces; linear restoring force; potential energy curve and equilibria – stable, unstable and neutral.

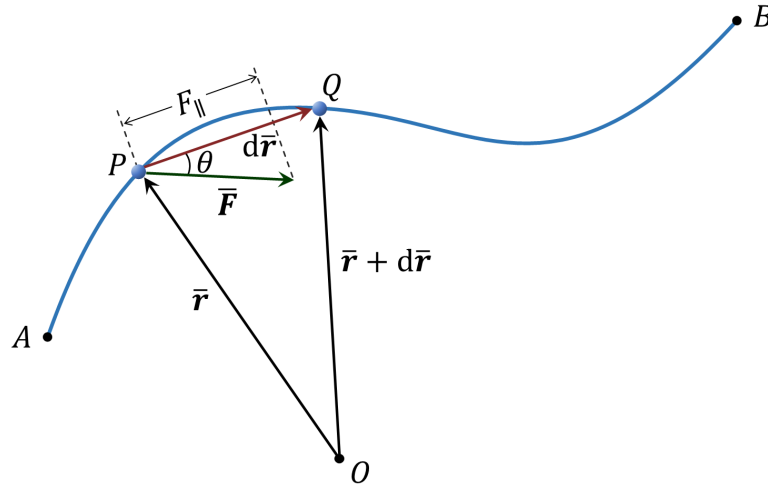
1 Concepts of work, energy and power

Before examining the work-energy theorem and conservation laws in mechanics, we must establish rigorous vector definitions for the foundational quantities of work, energy, and power:

- *Work*: an activity where a force acting on a system causes a displacement.
- *Energy*: the capacity of a system for doing work. A system may possess energy in the form of motion or position.
- *Kinetic energy*: the energy possessed by a system by virtue of its motion.
- *Potential energy*: the energy possessed by a system by virtue of its position or configuration.
- *Power*: the rate at which work is done or energy is used by a system.

1.1 Work

Let us consider a particle of mass m which is moving on an arbitrary path between points A and B . Suppose when the particle is at point P with a position vector $\mathbf{r}$, it is under the influence of some force $\mathbf{F}$ which causes it to get displaced to another point Q with a position vector $\mathbf{r} + d\mathbf{r}$.



The infinitesimal amount of work done by the force $\mathbf{F}$ in moving the particle from P to Q is

$$dW = \mathbf{F} \cdot d\mathbf{r} = F \cos \theta dr = F_{\parallel} dr$$

where θ is the angle between $\mathbf{F}$ and $d\mathbf{r}$ and $F_{\parallel} = F \cos \theta$ is the projection of $\mathbf{F}$ along $d\mathbf{r}$.

Hence, the total work done by the force $\mathbf{F}$ in moving the particle from A to B is

$$W = \int_A^B dW = \int_A^B \mathbf{F} \cdot d\mathbf{r} = \int_A^B F \cos \theta dr$$

1.2 Energy

1.3 Power

2 Work and energy

2.1 Work-energy theorem

2.2 Potential energy

3 Conservative and non-conservative forces

3.1 Conservative force as a negative gradient of potential energy

3.2 Law of conservation of mechanical energy

3.3 Work done by non-conservative forces

4 Linear restoring force

Insert section content.

5 Potential energy and equilibria

5.1 Potential energy curve

5.2 Types of equilibria

6 General law of conservation of energy

Insert content here.